Quinten Bubberman

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Fourth-year game programming student at Breda University of Applied Sciences specializing in graphics and engine systems programming. Experienced with Vulkan rendering pipelines, ECS architecture, and performance optimization. Focused on low-level systems, GPU-driven techniques, and building robust engine tools.

Skills

Programming languages: C++, Rust, Python, GLSL
APIs: Vulkan, EnTT, SDL3, GLFW
Engines: Unreal Engine, Godot Engine
Tools: Git, Perforce, Rider, Visual Studio, NSight Graphics, Tracy, Linux
Natural Languages: Dutch (native), English (fluent)

Projects

Radiance Cascades: 2D GI **Sep 2026 – Present**
C++ Graphics Programmer - Solo
Implemented global illumination with radiance cascades, using a jump flood SDF for ray marching, bilinear-fix cascade merging, and a Morton-ordered probe direction layout. Paintable emissive and solid materials. Instanced sprites cast shadows and receive light. Built on my own Vulkan render graph library.

Vulkan GPU-Driven Renderer **Nov 2025 – Mar 2026**
C++ Graphics Programmer - Solo
Implemented Multi-Draw Indirect with a compute shader generating draw commands from storage buffers, GPU frustum culling with plane extraction from the projection matrix, and a visibility buffer pre-pass storing triangle and draw IDs for compute shader attribute reconstruction. Rendering 636K objects at 60 FPS in a single CPU draw call.

Beetle: Custom Engine **Feb 2025 – Apr 2025**
C++ Engine & Tools Programmer - Team
Developed Blender pipeline with custom GLTF property exporting and Python tooling. Built ECS-based UI system with live editor, cubic Bezier spline system for AI pathfinding, spring arm camera, 3D audio with FMOD, and input mapping. Shipped on PC and Steam Deck.

Education

Breda University of Applied Sciences **Sept 2023 – Jul 2027**
Bachelor's, Game Technologies *Breda, Netherlands*
Nehalennia **Sept 2018 – Jul 2023**
HAVO (Senior general secondary education) *Middelburg, Netherlands*